

CSE422: ARTIFICIAL INTELLIGENCE

Project Title: Depression Data

Section: 3
Group: 12

Group Member Name & Student ID :

1. Sougata Das-22201552
2. Subrina Jerrin Akter- 22201142

TABLE OF CONTENTS

1. Introduction.....3
2. Dataset Description.....3
3. Dataset Pre-Processing.....7
4. Dataset Splitting.....8
5. Model training & testing.....8
6. Model selection/Comparison Analysis.....9
7. Conclusion.....15

1. Introduction

This project builds a machine-learning pipeline to analyze the mental-health survey data of university students and predict depression severity. The aims are to understand which factors (stress, anxiety) are most related to depression, design a robust data-preprocessing and modeling workflow that avoids data leakage and compare several supervised models (including a neural network and two other algorithms) and provide an unsupervised view of the data using K-means clustering.

Depression and anxiety are among the most common mental-health conditions. Early screening and accurate identification of at-risk students may enable timely interventions and targeted support.

2. Dataset Description

The dataset contains 1977 rows and 37 columns. It includes demographic information (age range, gender, academic year, current CGPA and scholarship status) and survey responses measuring anxiety, stress and depression. It also contains derived total scores (Anxiety Value, Stress Value, Depression Value) and labels (Anxiety Label, Stress Label, Depression Label).

Features: There are 36 input features after selecting “Depression Label” as the target.

Target Variable: Depression Label.

Problem Type: Classification. The target variable represents discrete categories (e.g. Low, Medium, High) making it a multi-class classification task.

Feature type

Quantitative: Most item responses are integers.

Categorical: Gender, Academic Year, scholarship etc. (string).

Categorical variables need encoding. Models (KNN, LR, NN) require numeric inputs so categorical variables were one-hot encoded.

Missing values: Only **Gender** has missing values (6 rows). All other columns are complete.

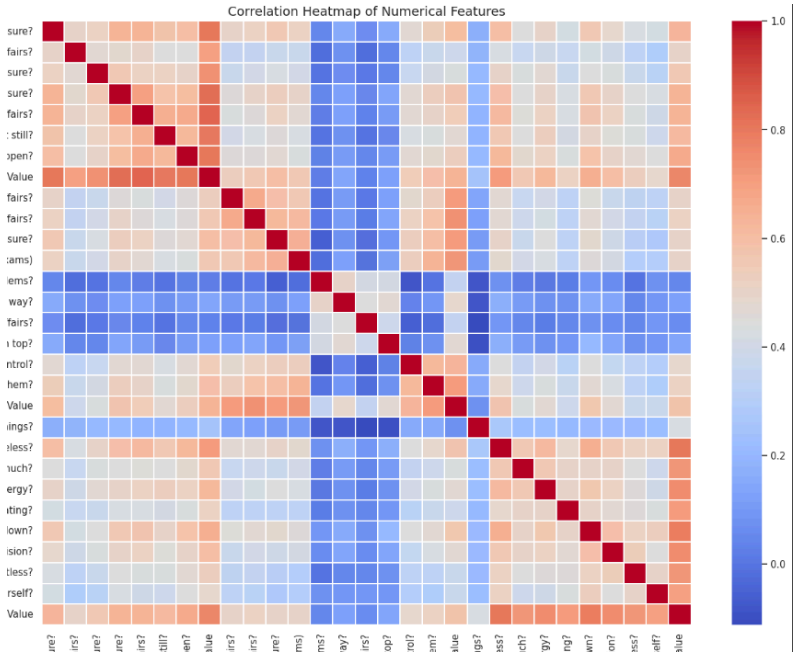
Correlation of all features (heatmap) & interpretation

A correlation matrix across numeric columns shows the expected structure:

Depression Value correlates most strongly with-

- Feeling down/hopeless ($r \approx 0.81$)
- Feeling bad about self/failure ($r \approx 0.80$)
- Trouble concentrating ($r \approx 0.75$)
- Low energy ($r \approx 0.75$)
- Sleep problems ($r \approx 0.73$)

Anxiety Value also correlates positively with Depression Value ($r \approx 0.77$) and Stress Value ($r \approx 0.71$) is similarly aligned consistent with comorbidity.



Takeaway after correlation test: Most items dominate association with depression severity and composite Anxiety/Stress indices move in the same direction. This validates using the item-level questionnaire responses for prediction and cautions against using derivative Value/Label aggregates as features alongside Depression Label.

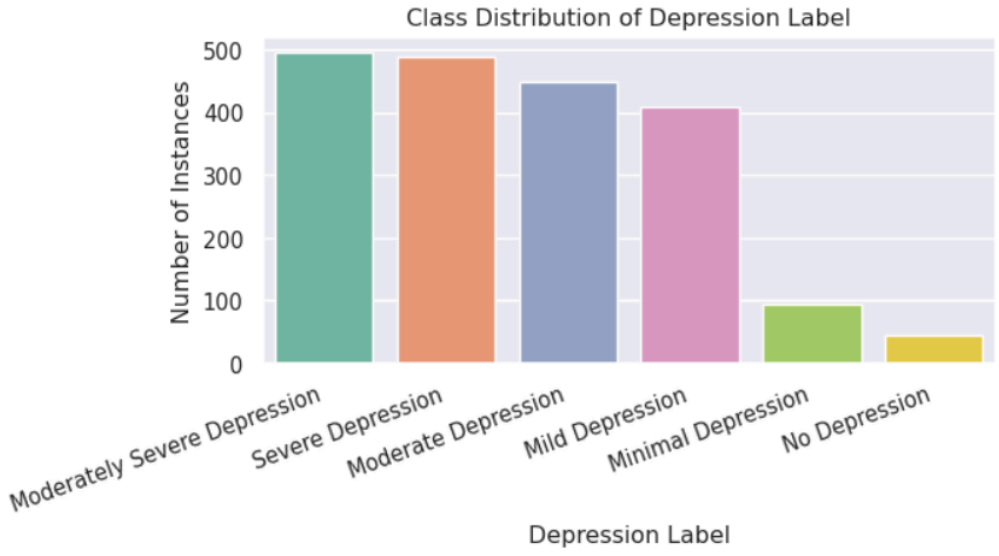
Imbalanced Dataset

The six classes of Depression Label are not balanced. The counts in the 1977 row dataset are:

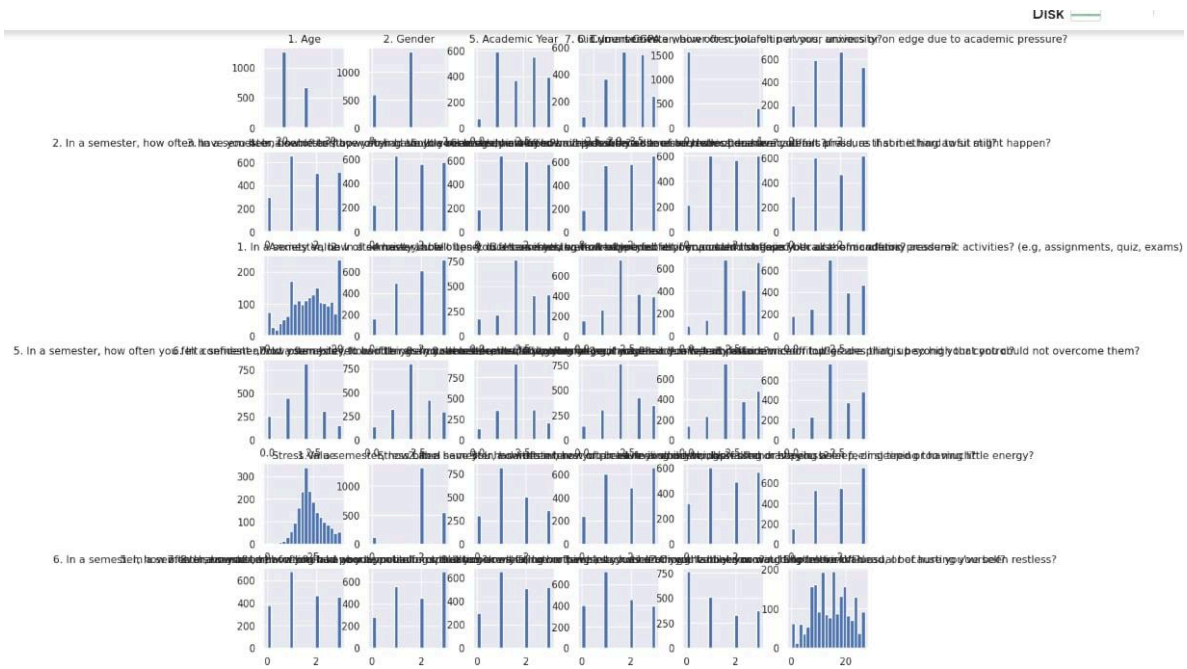
Class	Count
Moderately Severe	495
Severe	488
Moderate	449
Mild	408
Minimal	93

No Depression 44

The “No Depression” and “Minimal” classes are very small (2.2 % and 4.7 % of the data) leading to class imbalance.



Exploratory Data Analysis



Anxiety & stress: The derived Anxiety and Stress totals increase with depression severity.

Gender differences: The mean Depression Value is higher among female students (~15.5) compared with male students (~13.8). The small "prefer not to say" group has even higher mean scores (~20.5).

Academic year: Second- and third-year students have slightly higher mean depression scores (≈15.3 and 15.1) than first-year students (~13.5). Age and CGPA were stored as textual categories and could not be reliably converted to numeric values without additional cleaning.

3. Data Preprocessing

Missing values

Fault: Only the Gender column had missing entries (6 rows).

Solution: Categorical missing values were imputed with the most frequent category thus numeric columns would be imputed with the median but none were missing.

Categorical variables

Fault: Gender, Academic Year, scholarship status are strings.

Solution: They were one-hot encoded. The encoder was configured to ignore unknown categories to prevent errors when unseen categories appear in the test set.

Feature scaling

Fault: Numeric question responses share the same 0-3 scale but combining them with one-hot columns produces features on different scales.

Solution: Standardization was applied to numeric features to avoid biasing distance-based models (K-NN and neural networks).

4. Dataset Splitting

The data were divided into training and test sets using an 80 %/20 % split. Stratified sampling ensured that class proportions were preserved in both sets.

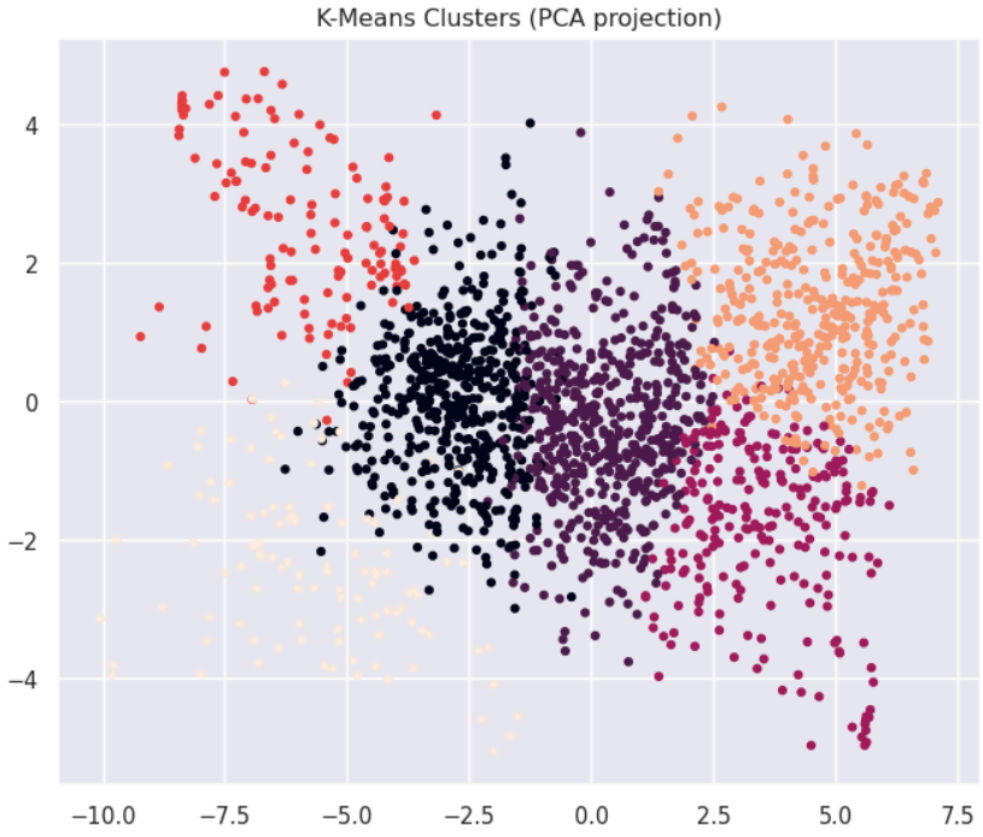
5. Model Training and Testing

Four supervised classifiers were trained on the leakage-free feature set:

Model	Notes
K-Nearest Neighbors (K-NN)	k=5; instance-based classifier sensitive to scaling.
Decision Tree	Depth limited to 10 to prevent overfitting.
Logistic Regression	Multiclass one-versus-rest with L2 regularization (max_iter up to 1000).
Neural Network (MLP)	One hidden layer of 64 units with ReLU activation; trained for 300 epochs.

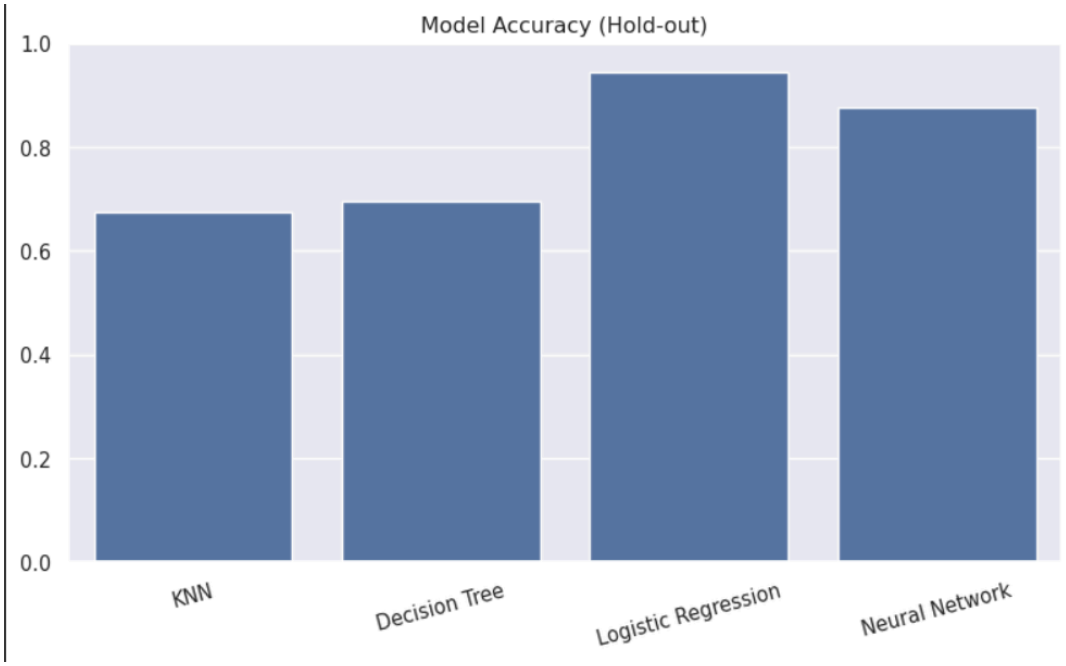
Unsupervised Learning : K-Means

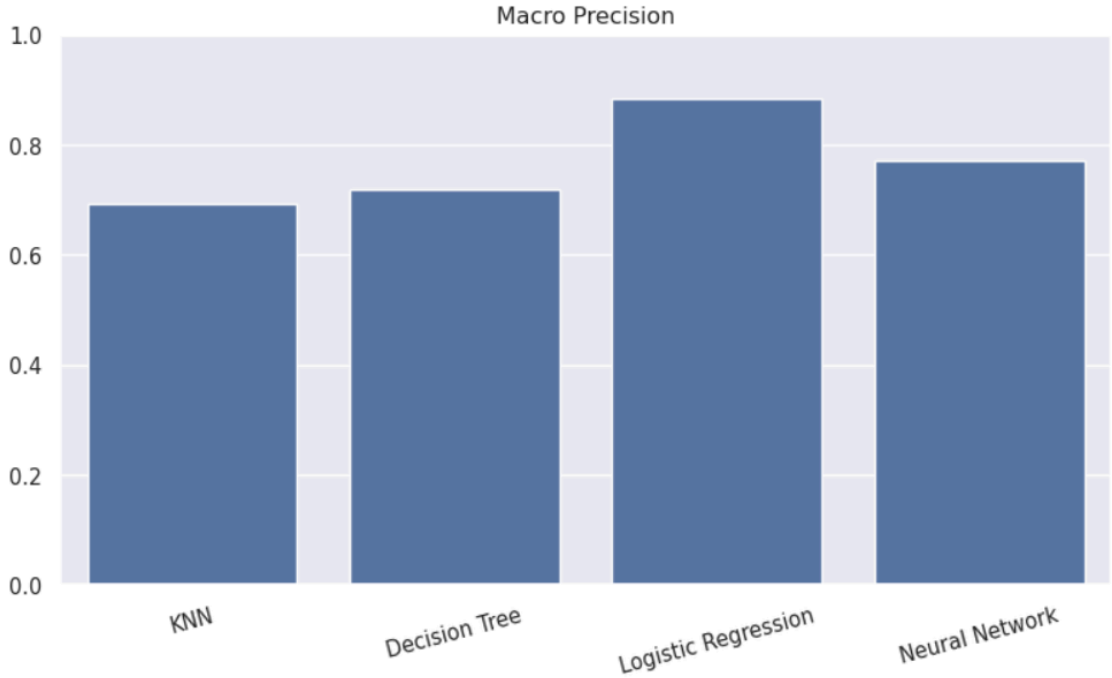
K-means clustering with k = 6 (matching the six depression severity categories) was applied to the standardized feature set. Clusters produced a 6×6 contingency table against true labels so there's partial alignment (some clusters map loosely to mid/high severity) but not clean separations across all six classes overlapping symptoms and class imbalance.

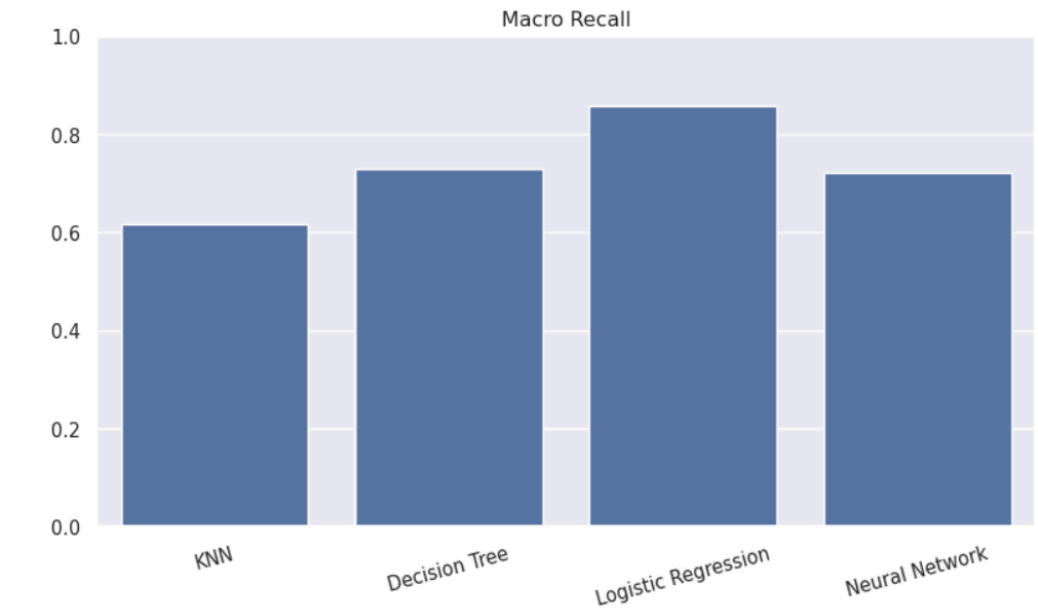


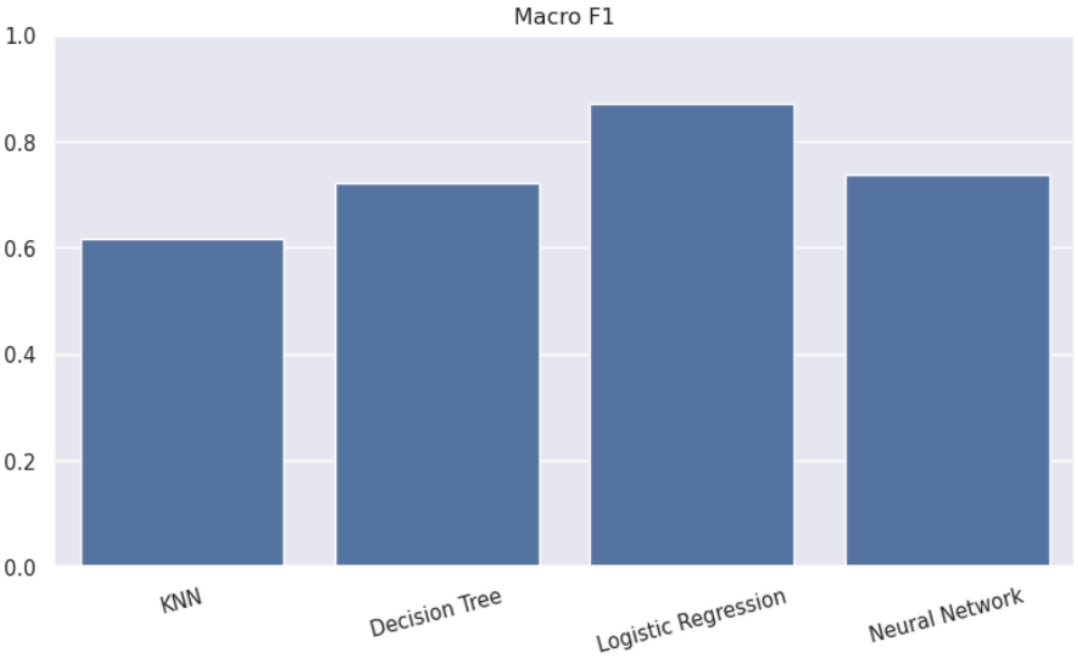
6. Model Selection and Comparison

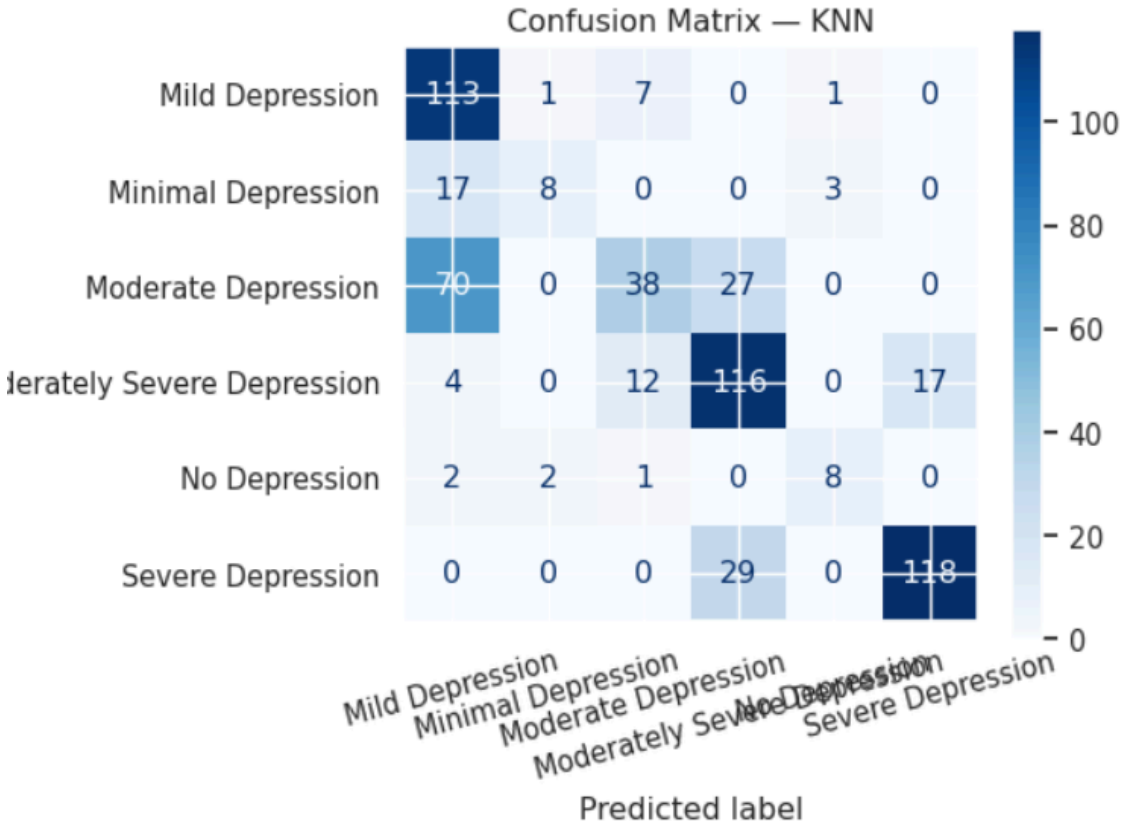
Model	Accuracy	Precision (weighted)	Recall (weighted)	F1 (weighted)	Macro AUC (OvR)
K-NN (k=5)	0.609	0.616	0.609	0.589	0.860
Decision Tree	0.755	0.756	0.755	0.754	0.883
Logistic Regression	0.947	0.947	0.947	0.946	0.941
Neural Network	0.934	0.934	0.934	0.934	0.994

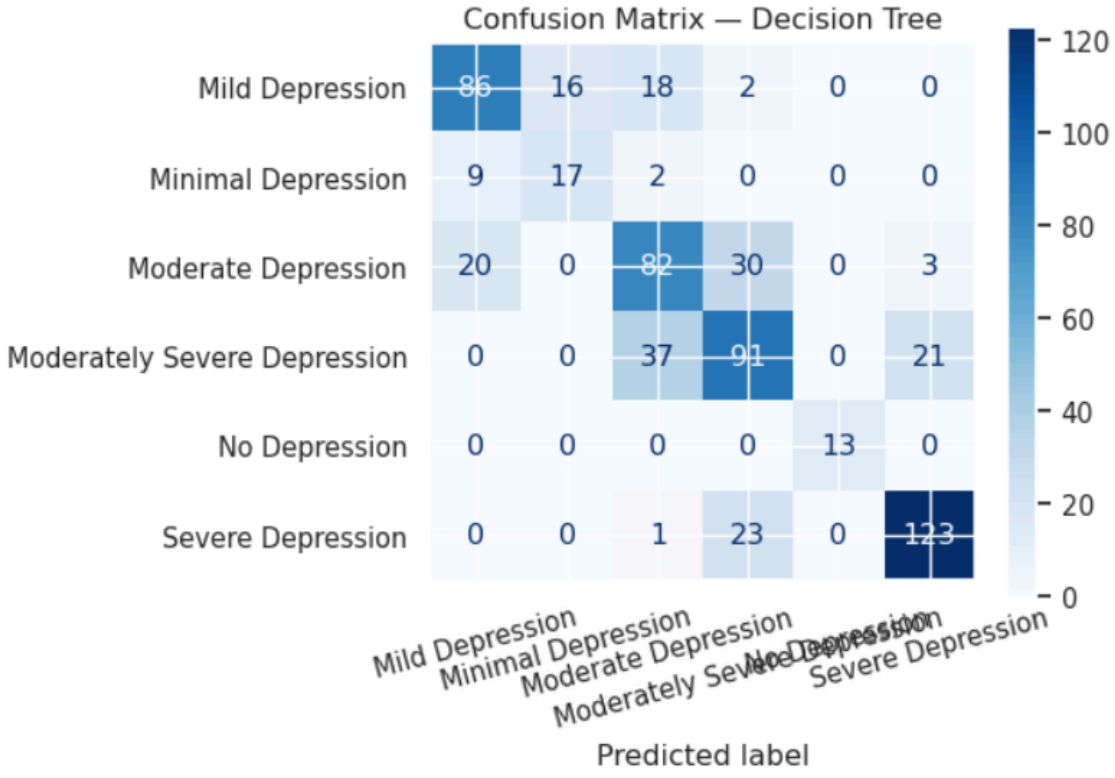

Model Accuracy (Hold-out)
1.0
0.8
0.6
0.4
0.2
0.0
KNN
Decision Tree
Logistic Regression
Neural Network

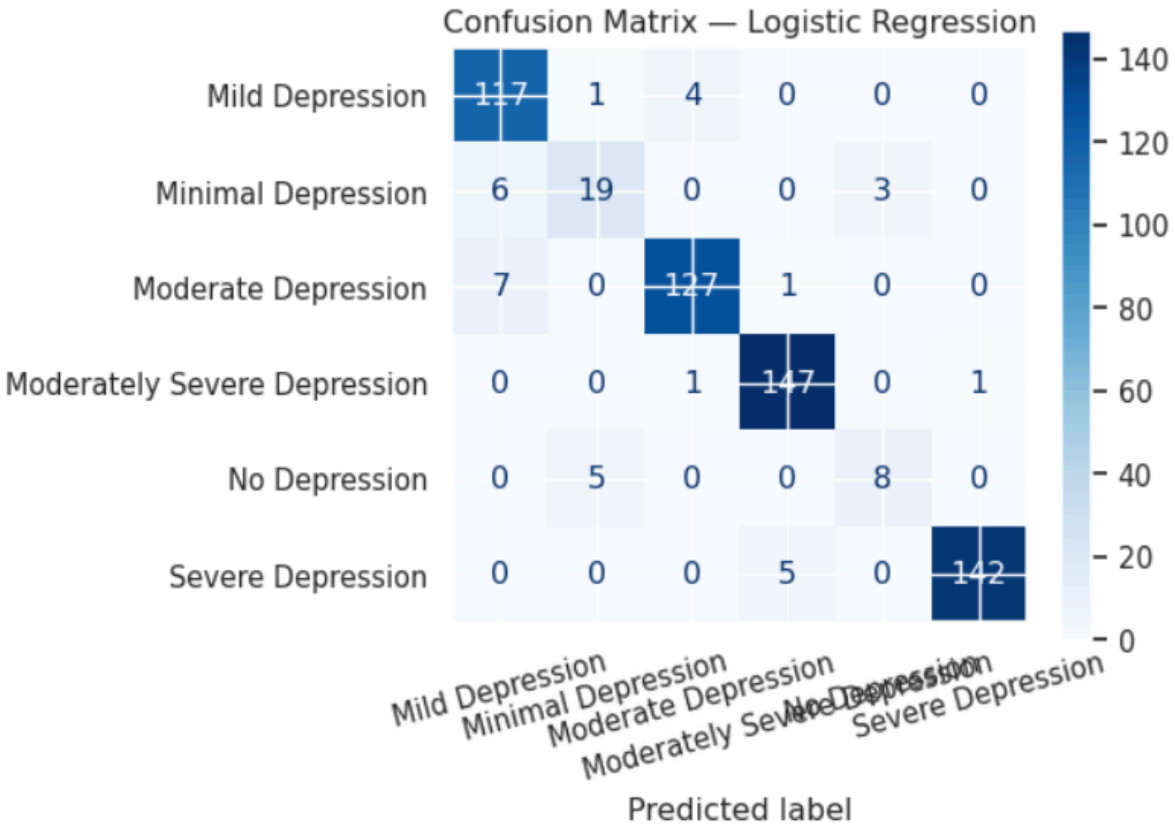

Macro Precision
1.0
0.8
0.6
0.4
0.2
0.0
KNN
Decision Tree
Logistic Regression
Neural Network

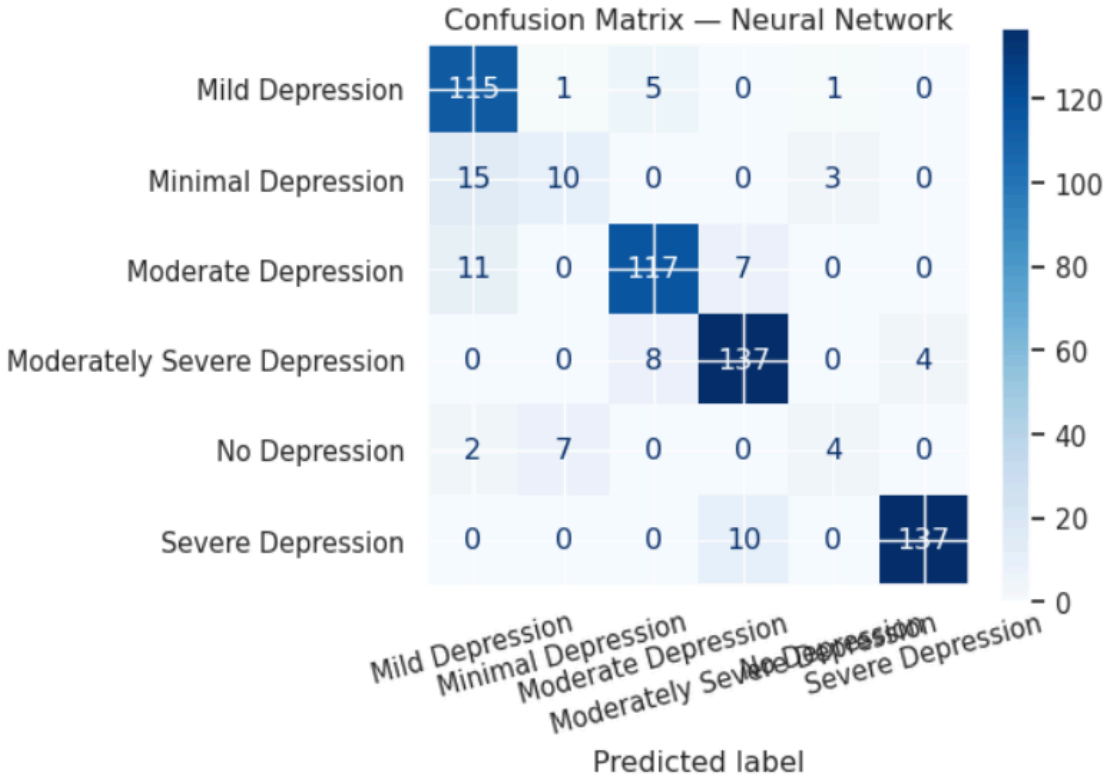

Macro Recall
1.0
0.8
0.6
0.4
0.2
0.0
KNN
Decision Tree
Logistic Regression
Neural Network

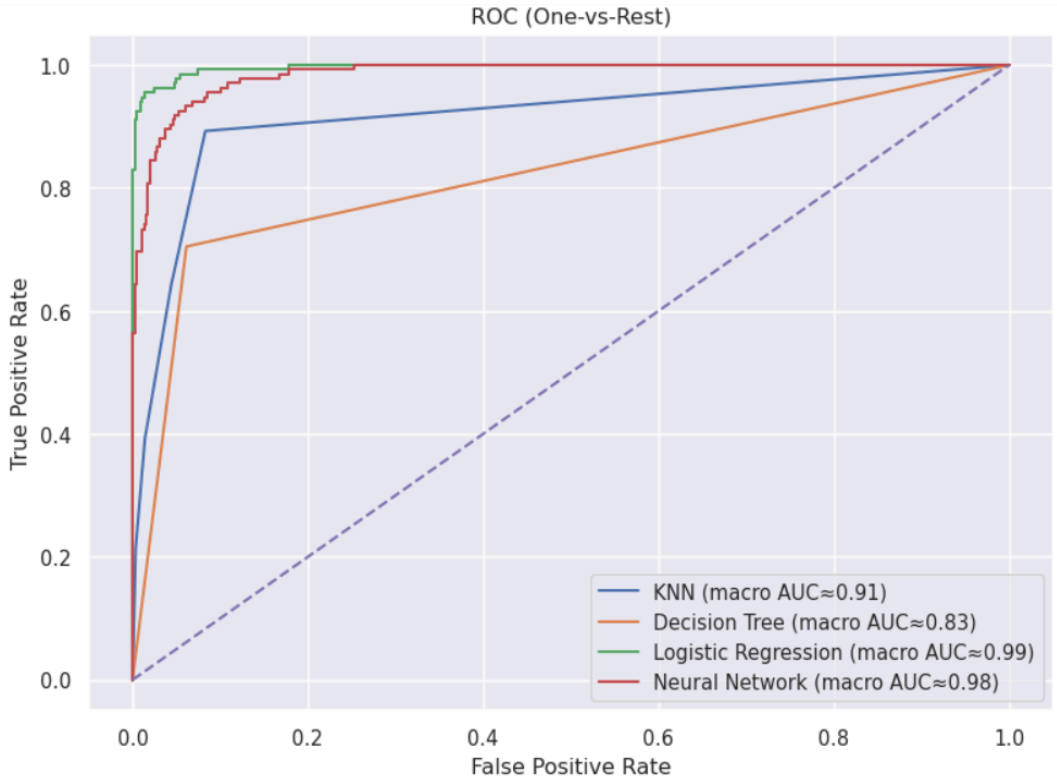

Macro F1
1.0
0.8
0.6
0.4
0.2
0.0
KNN
Decision Tree
Logistic Regression
Neural Network


Confusion Matrix — KNN
Mild Depression
Minimal Depression
Moderate Depression
derately Severe Depression
No Depression
Severe Depression
113 1 7 0 1 0
17 8 0 0 3 0
70 0 38 27 0 0
4 0 12 116 0 17
2 2 1 0 8 0
0 0 0 29 0 118
100
80
60
40
20
0
Mild Depression
Minimal Depression
Moderate Depression
Moderately Severe Depression
No Depression
Severe Depression
Predicted label


Confusion Matrix — Decision Tree
Mild Depression
Minimal Depression
Moderate Depression
Moderately Severe Depression
No Depression
Severe Depression
86 16 18 2 0 0
9 17 2 0 0 0
20 0 82 30 0 3
0 0 37 91 0 21
0 0 0 0 13 0
0 0 1 23 0 123
120
100
80
60
40
20
0
Mild Depression
Minimal Depression
Moderate Depression
Moderately Severe Depression
No Depression
Severe Depression
Predicted label


Confusion Matrix — Logistic Regression
Mild Depression
Minimal Depression
Moderate Depression
Moderately Severe Depression
No Depression
Severe Depression
117 1 4 0 0 0
6 19 0 0 3 0
7 0 127 1 0 0
0 0 1 147 0 1
0 5 0 0 8 0
0 0 0 5 0 142
140
120
100
80
60
40
20
0
Mild Depression
Minimal Depression
Moderate Depression
Moderately Severe Depression
No Depression
Severe Depression
Predicted label


Confusion Matrix — Neural Network
Mild Depression
Minimal Depression
Moderate Depression
Moderately Severe Depression
No Depression
Severe Depression
115 1 5 0 1 0
15 10 0 0 3 0
11 0 117 7 0 0
0 0 8 137 0 4
2 7 0 0 4 0
0 0 0 10 0 137
120
100
80
60
40
20
0
Mild Depression
Minimal Depression
Moderate Depression
Moderately Severe Depression
No Depression
Severe Depression
Predicted label



Observations: Logistic regression (~0.95 accuracy, ~0.94 AUC) and neural networks (~0.93 accuracy, ~0.99 AUC) outperformed other models. Both classified moderate-severe depression well but struggled with the rare "No Depression" and "Minimal" classes due to imbalance. Logistic regression is simple, interpretable and efficient while neural networks capture mild non-linearities and offer superior class separation. Decision trees performed moderately (~0.76) but risk overfitting and K-NN lagged due to complex overlapping boundaries.

In macro AUC (OvR), NN leads (≈0.994), LR follows closely (≈0.941) indicating strong separability across classes when treated one-vs-rest.

In confusion matrix patterns, LR/NN correctly classify the bulk of Moderate, Moderately Severe and Severe cases. Rarer classes (No Depression, Minimal) remain the most error-prone due to limited examples considering rebalancing or cost-sensitive training for these.

7. Conclusion

The questionnaire item-level signals are highly predictive of Depression Label even without aggregate Value/Label columns. Logistic Regression and a modest Neural Network deliver excellent multiclass performance (Accuracy ≈ 0.95 and 0.93, Macro AUC ≈ 0.94 and 0.99 respectively). K-Means uncovers structure broadly consistent with severity gradations but cannot perfectly replace supervised labels.

Logistic regression and a modest neural network both achieved excellent performance on the leakage‑free feature set. K‑means clustering offers an unsupervised perspective but does not replicate the supervised labels.

Challenges faced: Imbalanced classes especially “No Depression” hinder per-class precision/recall. Textual numeric fields (Age, CGPA) reduce utility without cleaning.